INTERNATIONAL SEMANTIC WEB CONFERENCE 2026 · BARI, ITALY

Does SPARQL federation work in the real world?

A longitudinal case study over large biological SPARQL endpoints

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[2] SIB Swiss Institute of Bioinformatics, Lausanne, Switzerland

[3] Swiss-Prot group, SIB Swiss Institute of Bioinformatics, Geneva, Switzerland

28 October 2026

SLIDES LINK

OUTLINE

Our five stage journey into the wild.



Motivation

Federation is tested on synthetic benchmarks

01–02

Our Design

67 real queries, 20+ live endpoints, 12+ months

03–05

Finding 1

Algorithmic federation succeeds far less often

06–07

Finding 2

Both approaches progressively worsen

08–09

Takeaways

For maintainers, users, and the community

10–13

⚠ Viewer discretion advised

This journey into the wild may, at times, prove unsettling. Rest assured: only some endpoints were harmed during this assessment, and all have since made a full recovery. Remedies for any frights encountered along the way are prescribed at the end.

One question, multiple databases, no centralization.

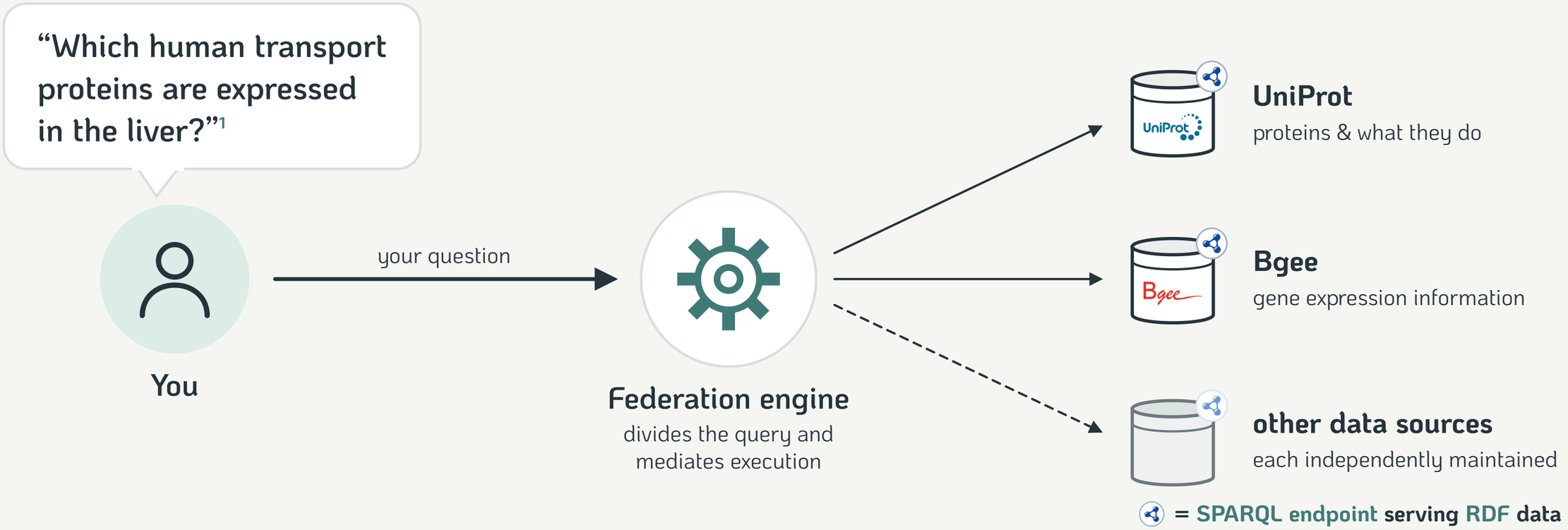
“Which human transport proteins are expressed in the liver?”[1]



You

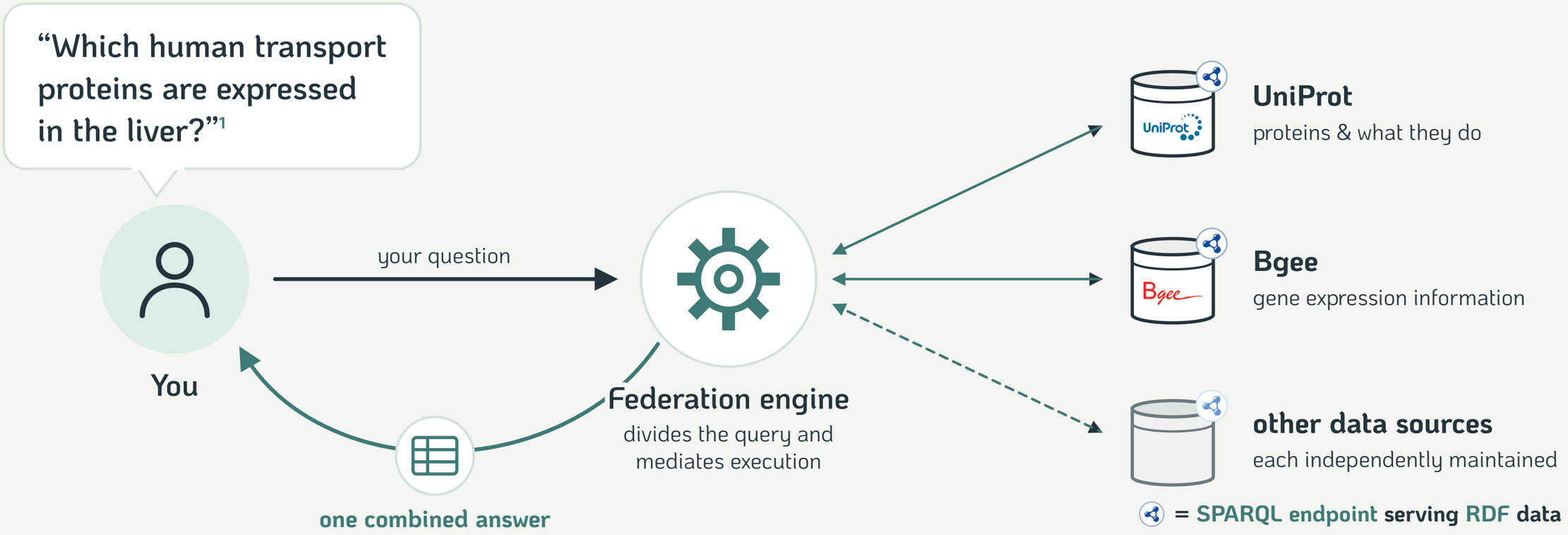
[1] Paraphrased from *query 109* which combines UniProt and Bgee.

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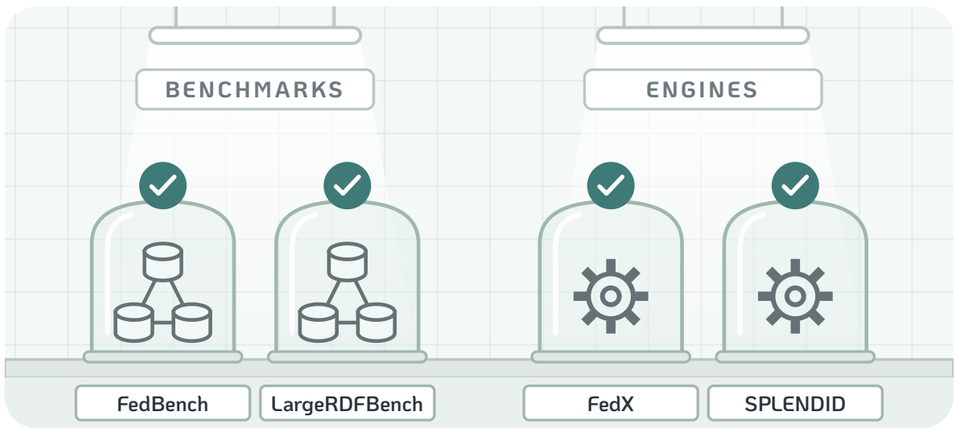
KEY MESSAGE Federation queries data where it lives and joins the results, without a central copy.

[1] Paraphrased from *query 109* which combines UniProt and Bgee.

Federation works in the lab. Does it work in the wild?

In the lab

prior benchmarks & engine papers[2]



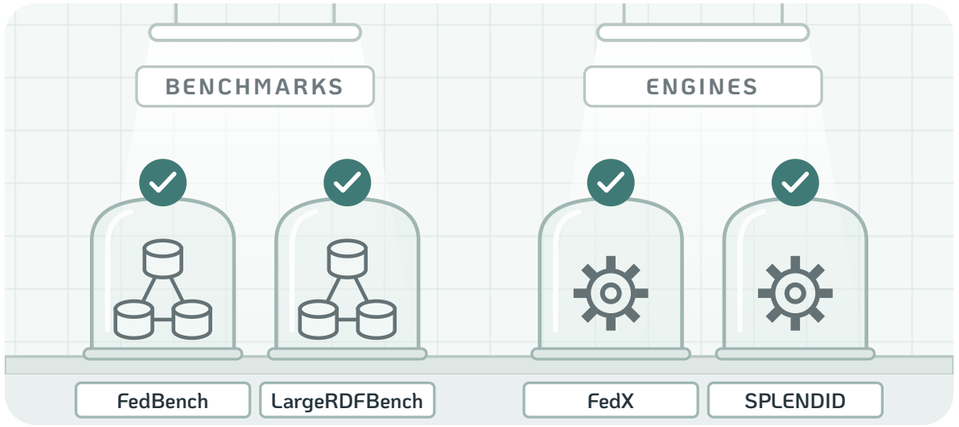
controlled · stable · reproducible

[2] Benchmarks: FedBench · LargeRDFBench; engines: FedX · SPLENDID; see also FedShop · Saleem et al. 2016 · Hernández et al. 2016

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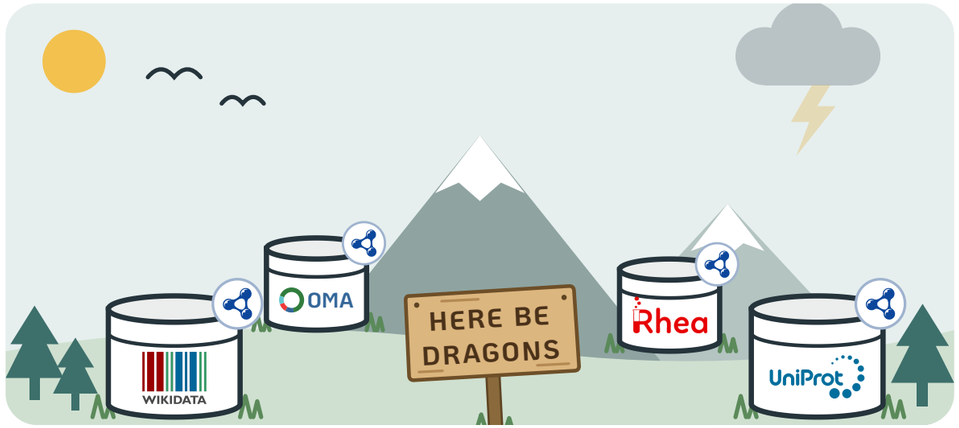


controlled · stable · reproducible

?

In the wild

everyday use, untested

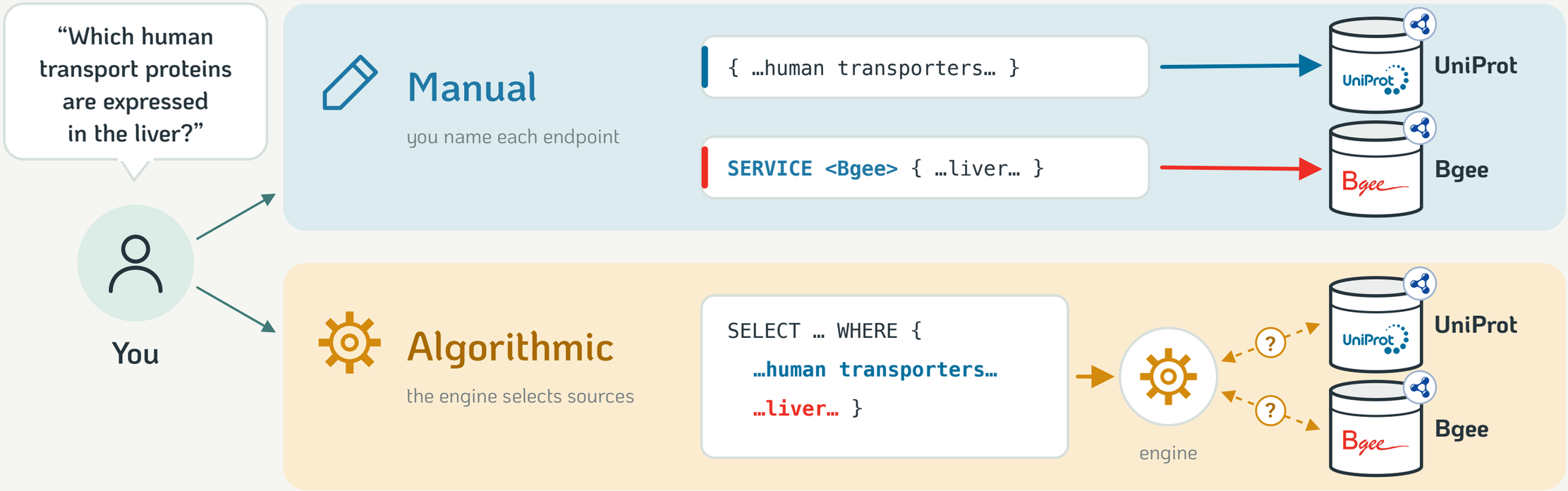


dynamic · unpredictable · untested

KEY MESSAGE Benchmarks test controlled conditions; we tested federation in everyday use.

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Source selection approaches:



Federation following user-defined `SERVICE` clauses

Manual

user defines query decomposition and endpoint targets with `SERVICE` clauses

human transporters

expressed in the liver

```
# + 8 PREFIX declarations
SELECT ?rhea ?protein ?anat
WHERE {
  GRAPH <https://sparql.rhea-db.org/rhea> {
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target





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target

1 whole query





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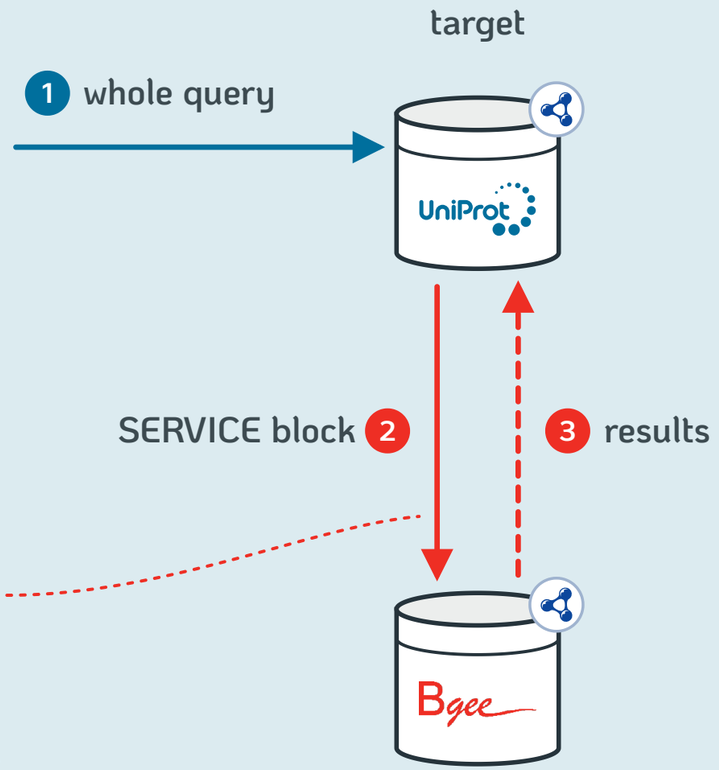
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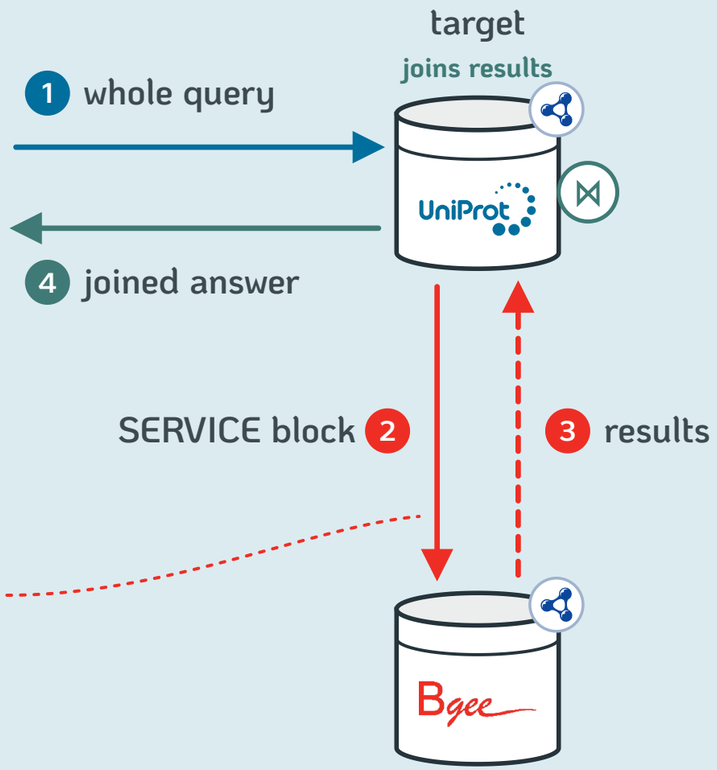
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Federation via a federation engine

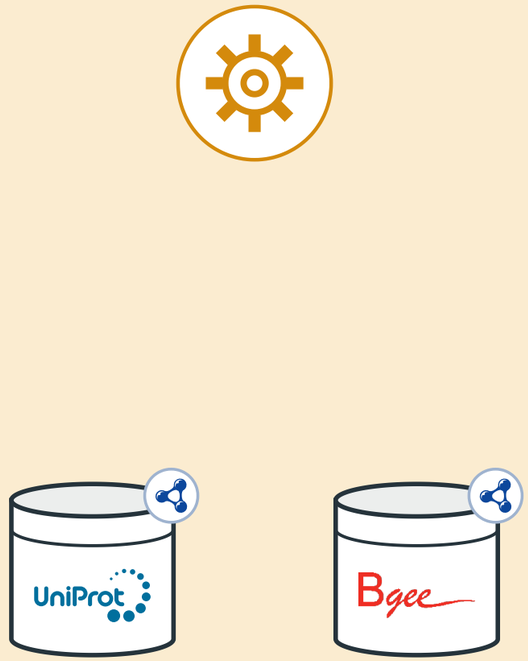
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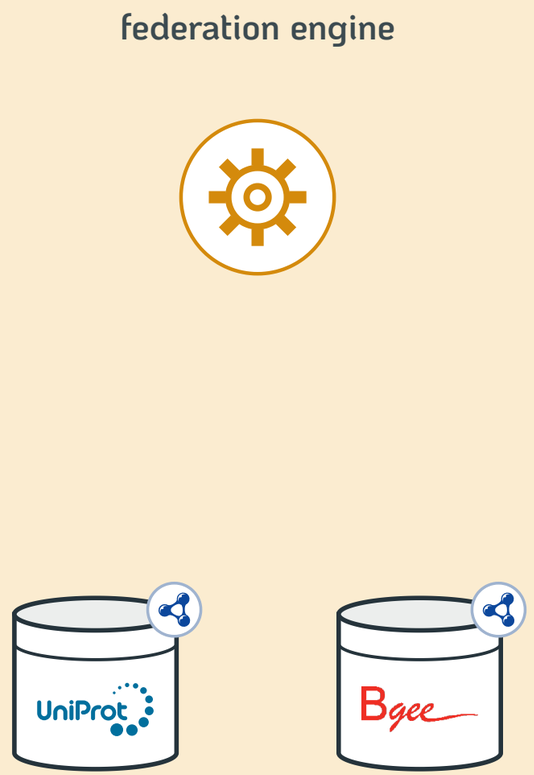
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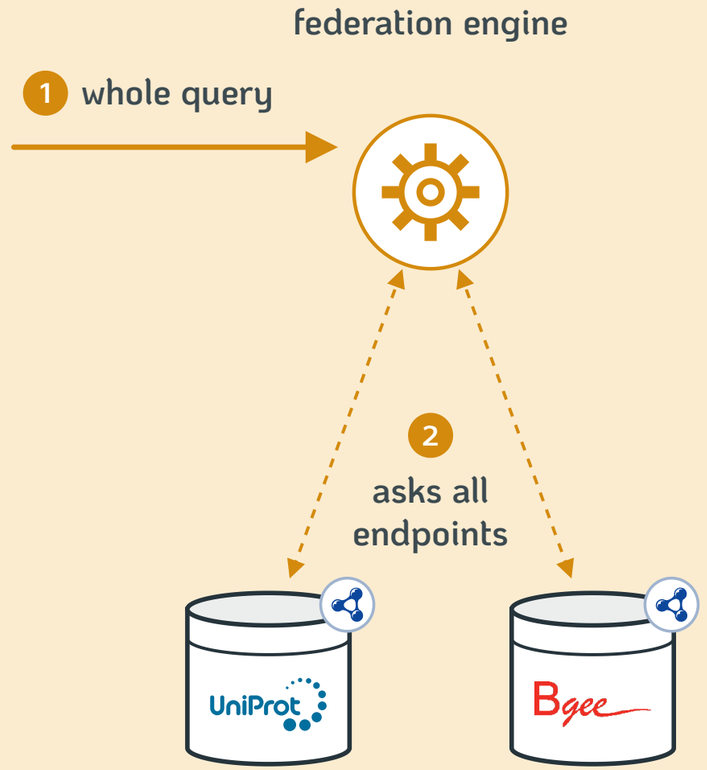
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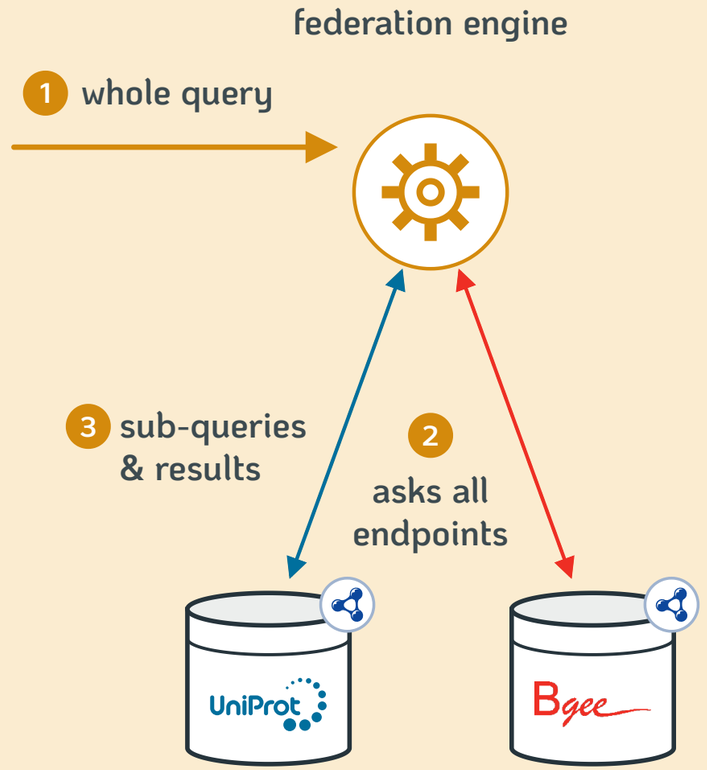
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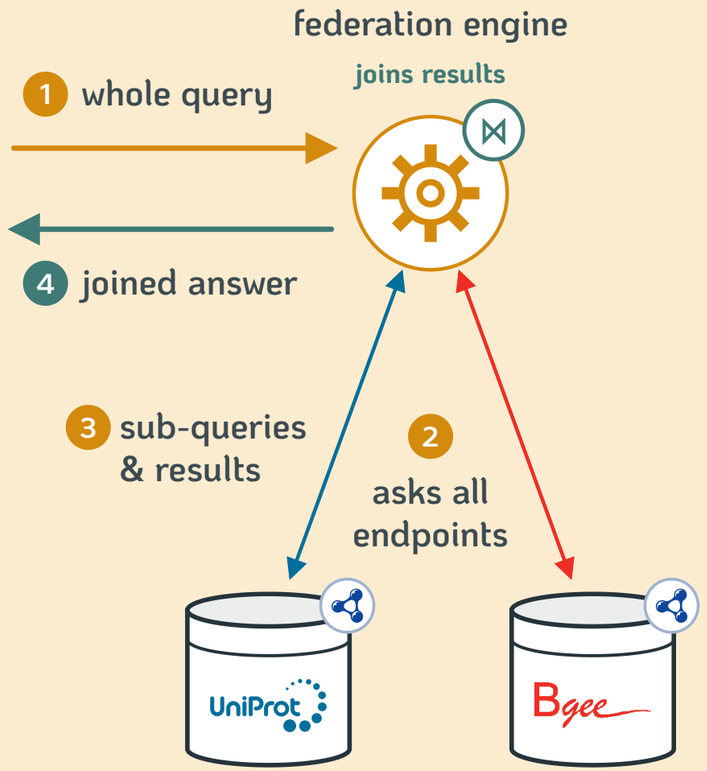
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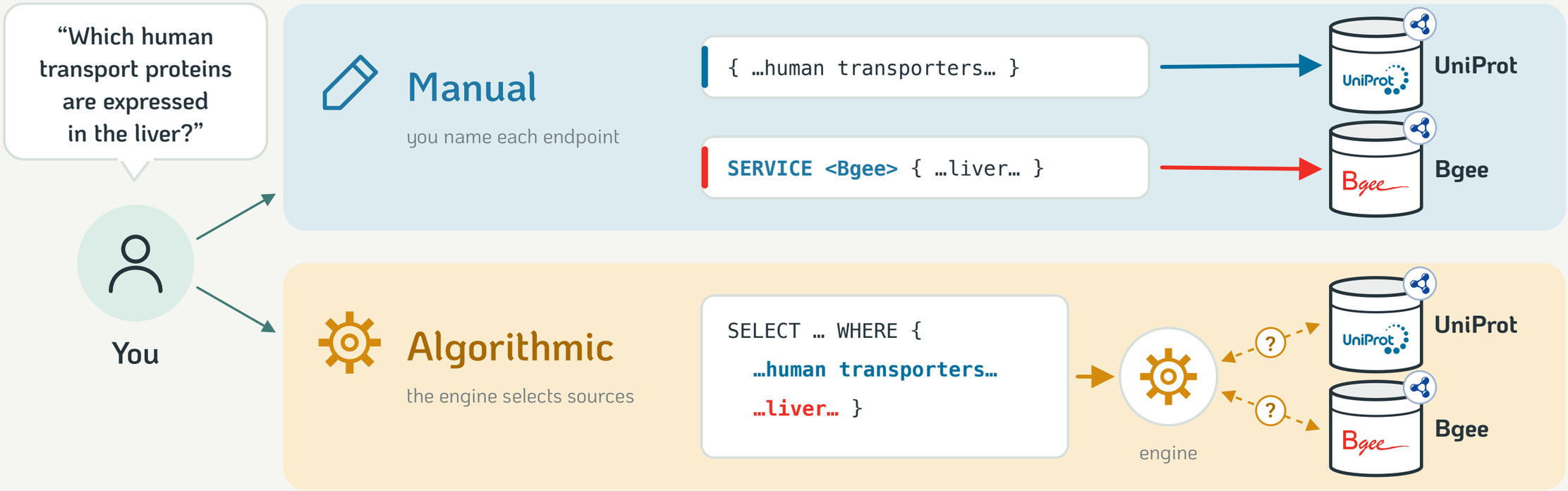
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[3] Run with Comunica using ASK-, COUNT- and VoID-based source selection; see S4.

Source selection approaches:



KEY MESSAGE Both approaches can theoretically provide the same answers to the same queries.

Real research questions, written by domain experts.

Domain experts
using SIB's public endpoints

→

Question + query pairs
published as endpoint examples

→

67 **federated**
queries in our study

Per-query complexity: S5 · General Query Statistics ↗ in the results explorer.

Real research questions, written by domain experts.

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67 **federated**
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What they ask (paraphrased)

"Which human cancer genes have counterparts in the rat brain?"
UniProt · OMA · Bgee

"Which protein patents have already expired?"
UniProt · European Patent Office

"Which human transport proteins are expressed in the liver?"
UniProt · Bgee · **our running example**

Why "real world"?

> ½ confirmed use or open question

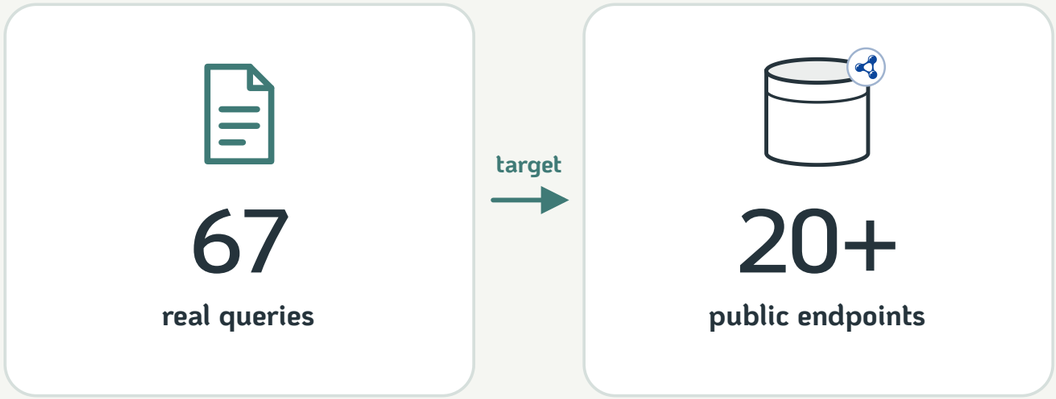
≈ ¼ results in peer-reviewed papers

2–3 sources, ≈ 15 triple patterns

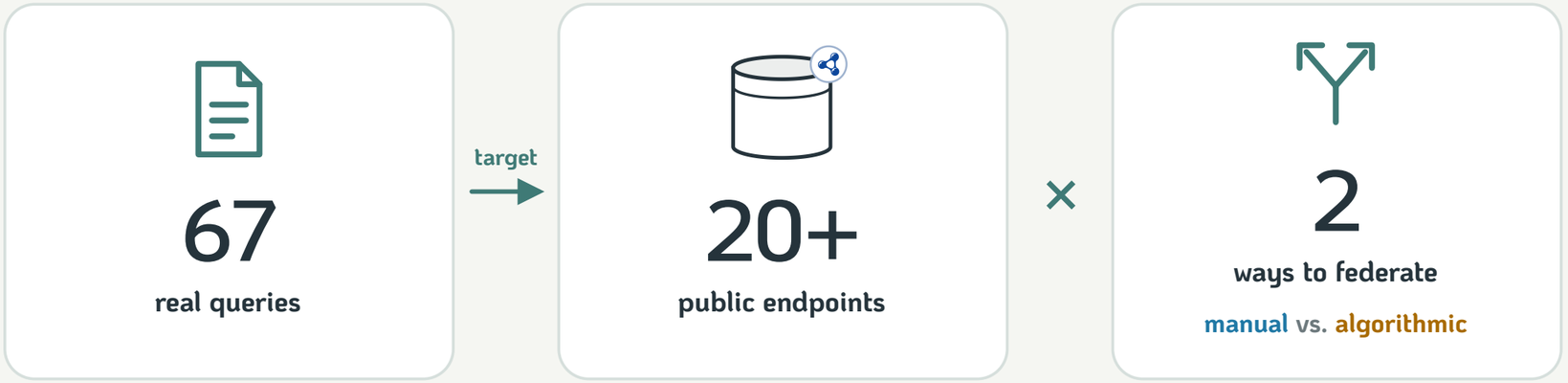
KEY MESSAGE Every query answers a real biological question, written by a domain expert.

Per-query complexity: S5 · General Query Statistics ↗ in the results explorer.

The same real-world queries, repeated over one year.



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The same real-world queries, repeated over one year.



1 run = every query executed once · 25 runs (4 manual, 21 algorithmic) · Mar 2025 – Jun 2026, shown in 4 periods

The same real-world queries, repeated over one year.

67
real queries

target →

20+
public endpoints

×

2
ways to federate
manual vs. algorithmic

×

25
runs over 14 months

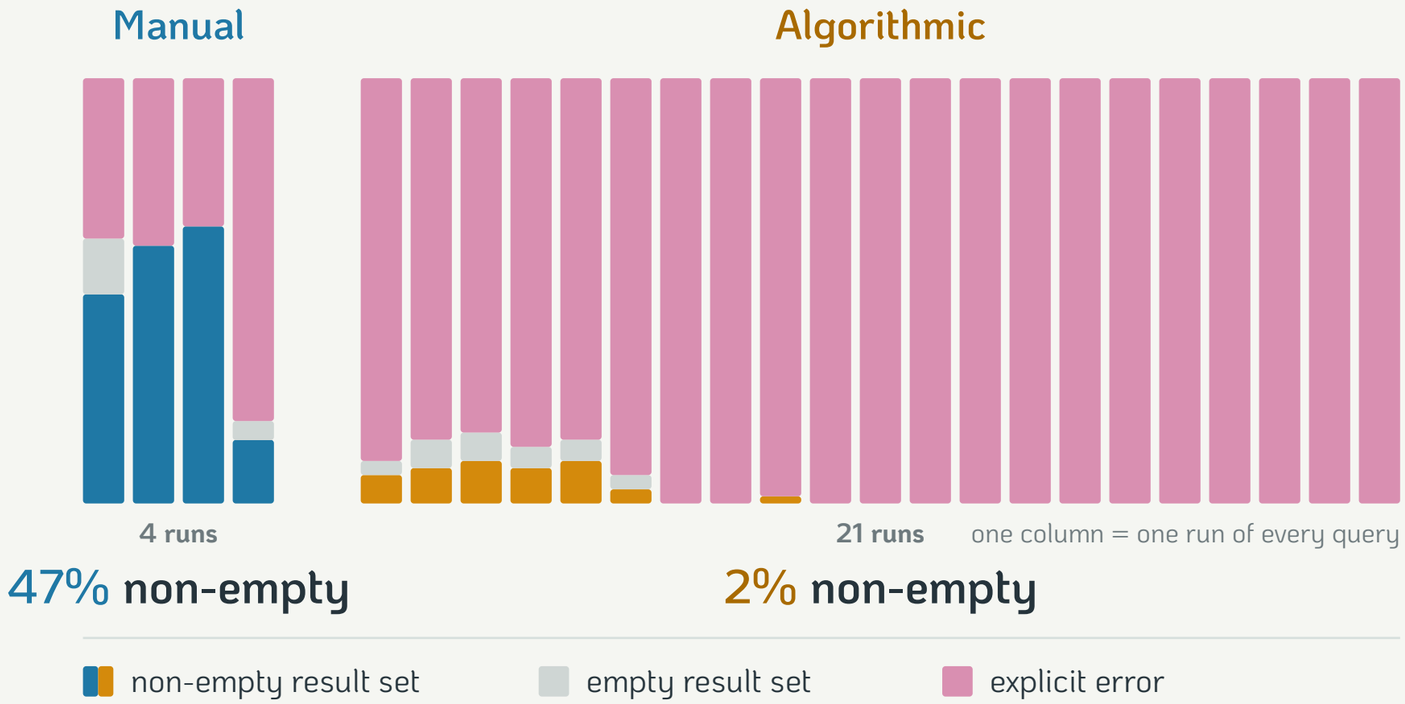
Spring 2025 — Fall 2025 — Winter 2026 — Spring 2026

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KEY MESSAGE A longitudinal study on live endpoints, i.e. federation as users actually experience it.

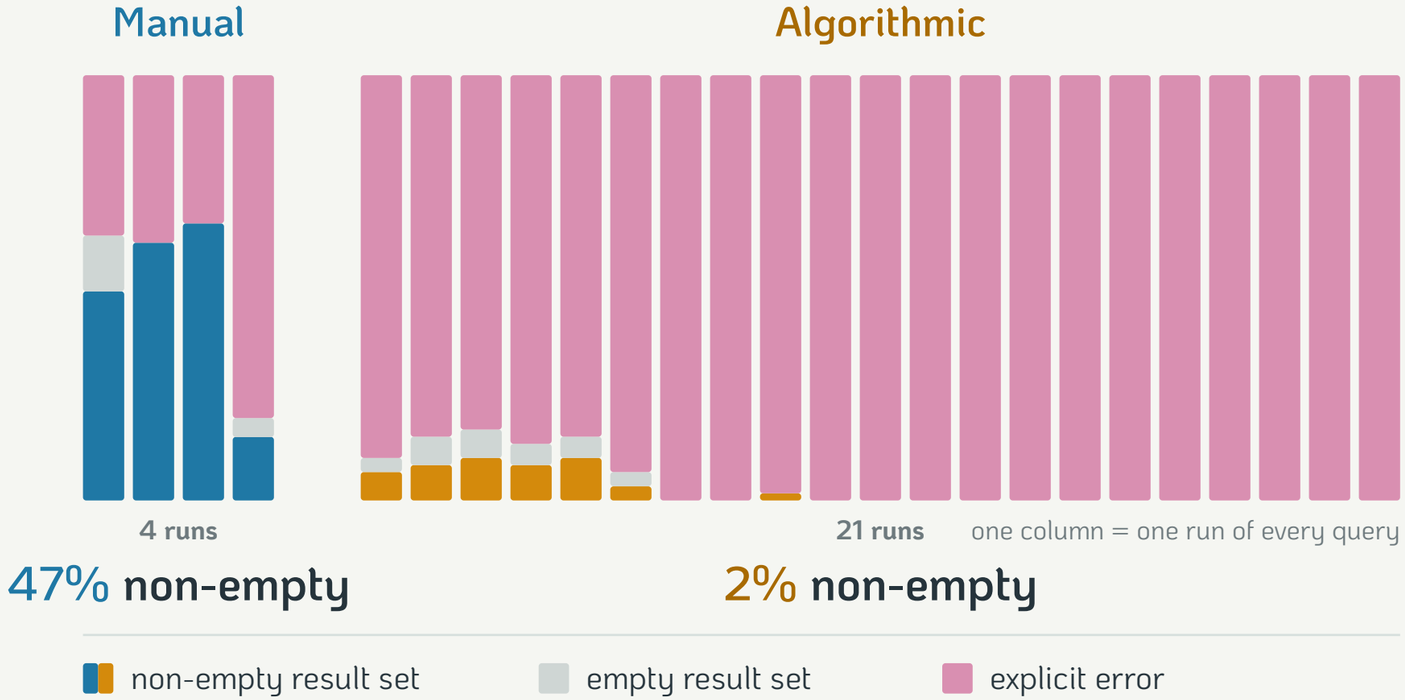
Explore every query and run in the interactive results explorer **ecrum19.github.io/fed-survey-results** ↗

Algorithmic source selection rarely produced results.



All 25 runs, Mar 2025 – Jun 2026: **Manual** 47.3% non-empty / 48.5% explicit errors (260 executions) · **Algorithmic** 2.3% / 96.3% (1,260 executions).
In Spring 2026, when both ran: **10** vs. **0** of 67 queries returned a non-empty result set.

Algorithmic source selection rarely produced results.



≈ 21×

lower success rate using **algorithmic** federation

96%

of **algorithmic** executions ended in an explicit error

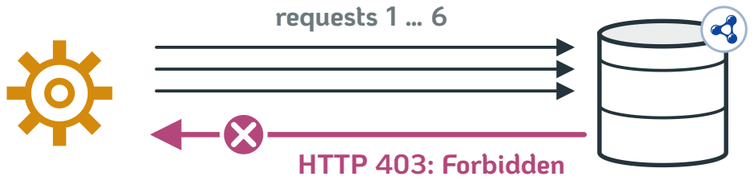
KEY MESSAGE **Algorithmic** federation was far less reliable than **manual** federation.

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Algorithmic probing exceeds endpoint usage limits.

Requests rejected early

54 queries · all 21 algorithmic runs



3–6 requests

before the endpoint starts replying **HTTP 403: Forbidden**

Request explosion

query 6 · one run



924,299 requests

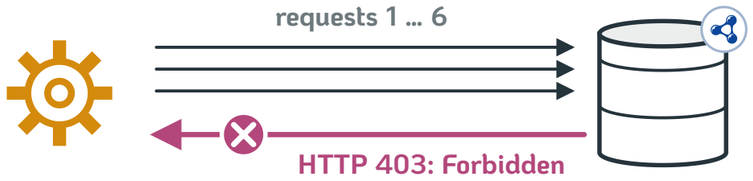
for a single query, **without a result**

3–6 requests: 533 of 814 algorithmic executions rejected with HTTP 4xx, mostly UniProt HTTP 403: Forbidden, Sep 2025 – Jun 2026; error groups: S6.
924,299 HTTP requests were issued by one algorithmic run of *query 6* before it failed (S7; results explorer ↗).

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Classification of the 1,027 explicit errors

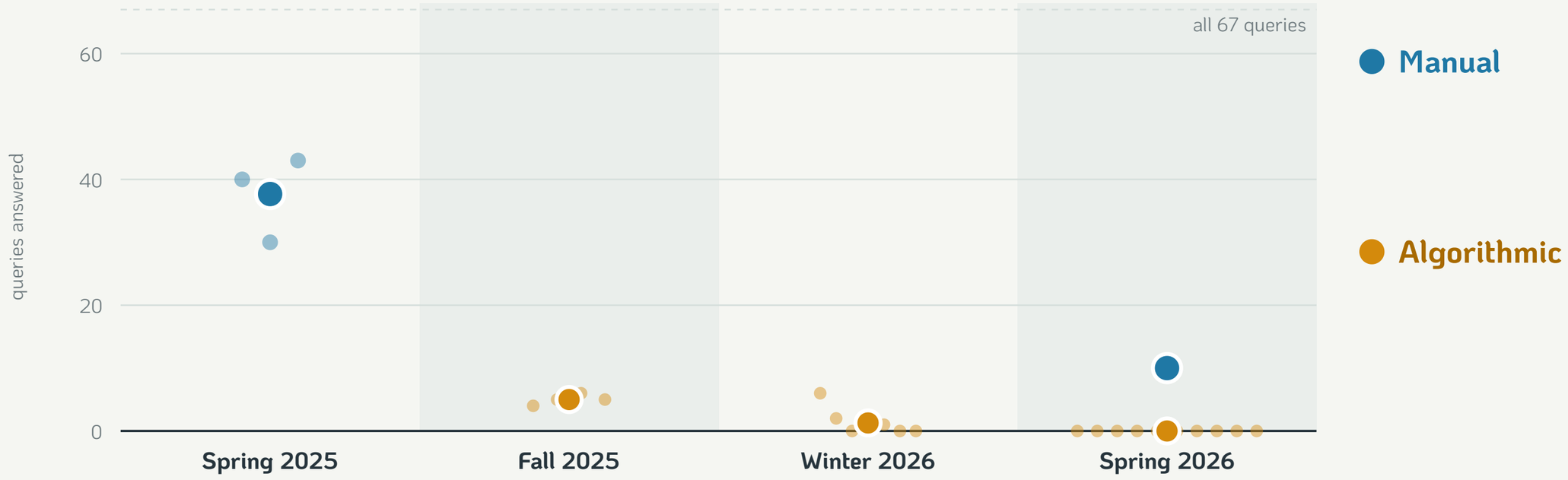


Rejected requests (HTTP 4xx) | Server errors (HTTP 5xx) | Other (network, parsing, time-outs)

KEY MESSAGE **Algorithmic** federation can exceed what live public endpoints are able to serve.

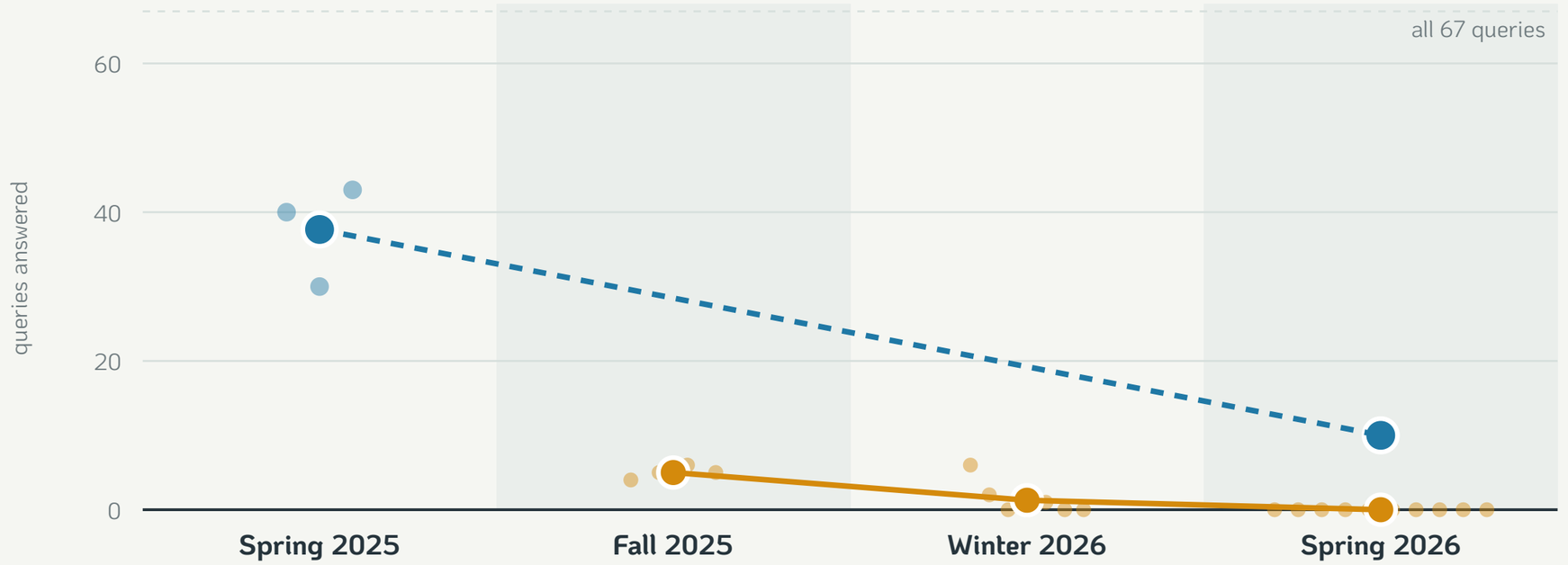
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Over 14 months, the share of answered queries declined.



Small dots = individual runs (25); large dots = period average.
Manual runs took place in Spring 2025 and Spring 2026; algorithmic runs from Fall 2025 on.
Per-run figure: S8.

Over 14 months, the share of answered queries declined.



Manual

38 → 10

Spring 2025 → Spring 2026

Algorithmic

5 → 0

Fall 2025 → Spring 2026

KEY MESSAGE Success declined over time, for both **manual** and **algorithmic** federation.

Small dots = individual runs (25); large dots = period average.
Manual runs took place in Spring 2025 and Spring 2026; algorithmic runs from Fall 2025 on.
Per-run figure: S8.

Public endpoints evolve as demand grows.

OBSERVED LIMITATIONS OF PUBLIC FEDERATION · MAR 2025 – JUN 2026

Endpoints move

DBGI/EMI moved to the METRIN-KG endpoint URL

14 of 67 queries affected

Services retire

BioSODA's temporary federation server, deprecated Sep, removed Nov 2025

14 of 67 queries affected

Limits tighten

UniProt traffic rose abruptly (likely AI crawlers[4]), resulted in stricter limits

requests rejected: HTTP 403

68.7% of queries showed result-count variation across repeated runs, including switches between returning results and returning none (paper, Sec. 5.1).
Details on each change: S9; the 14 affected queries: S10. [4] Suspected, not proven: R. Taelman, "Did AI Crawlers Kill SPARQL Federation?" (blog, 13 Apr 2026).

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2 in 3 | **queries returned a different number of results from run to run.**

Same query, same configuration: results were not reproducible.

Result heatmap ↗

KEY MESSAGE Endpoint changes broke working queries; federation clients could not adapt.

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Users need answers; maintainers need sustainability.



Query users

- Complex, multi-source questions
- Reliable answers
- Reproducible results



Endpoint maintainers

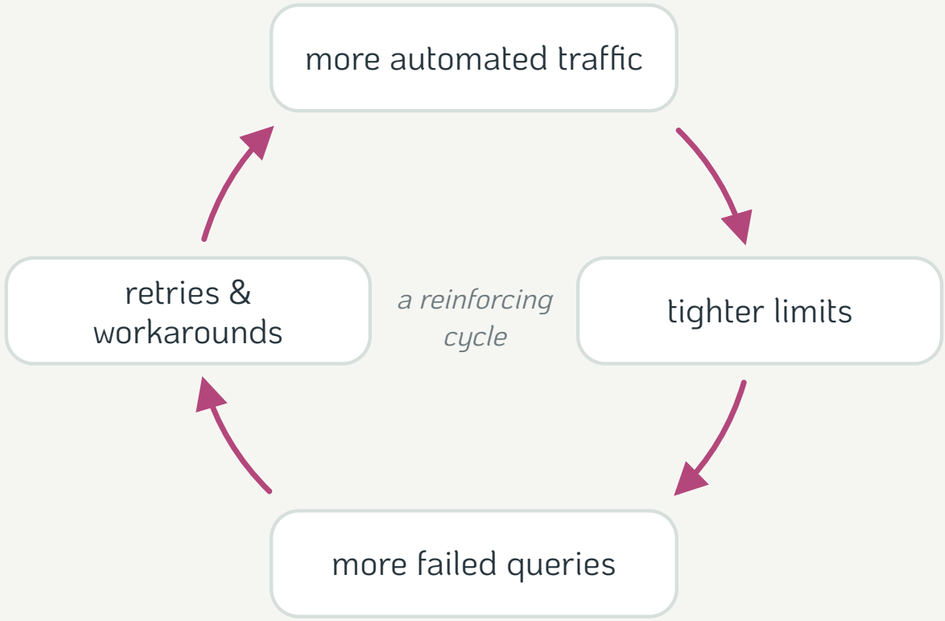
- Limited hardware and funding
- Fair access for all users
- Sustained service availability

Users need answers; maintainers need sustainability.



Query users

- Complex, multi-source questions
- Reliable answers
- Reproducible results





Endpoint maintainers

- Limited hardware and funding
- Fair access for all users
- Sustained service availability



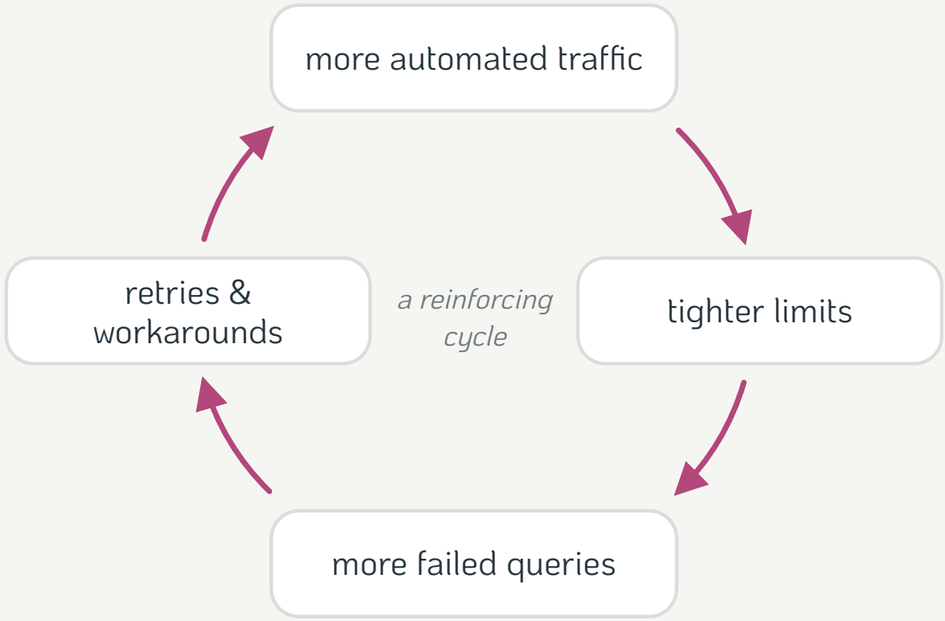
REAL EXAMPLE **UniProt** updated its usage limits as traffic grew; federation clients then received **HTTP 403** (**S9**).

Users need answers; maintainers need sustainability.



Query users

- Complex, multi-source questions
- Reliable answers
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Endpoint maintainers

- Limited hardware and funding
- Fair access for all users
- Sustained service availability



REAL EXAMPLE **UniProt** updated its usage limits as traffic grew; federation clients then received **HTTP 403** (**S9**).

KEY MESSAGE Both needs are legitimate; what is missing is coordination between them.

Reliable federation requires coordination.



Endpoint maintainers

→ **Publish the limits**
(in machine-readable form[5])



Engine developers

→ **Plan within the limits**
(request budgets, back-off, caching)



Researchers

→ **Test on live endpoints**
(repeatedly, over months)

[5] Existing starting points: VoID · SPARQL 1.1 Service Description

Reliable federation requires coordination.



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→ **Publish the limits**
(in machine-readable form[5])



Engine developers

→ **Plan within the limits**
(request budgets, back-off, caching)



Researchers

→ **Test on live endpoints**
(repeatedly, over months)

Shared, machine-readable agreements

what each endpoint allows
rate limits · time-outs · result caps

⟷

what each engine will send
request budget · back-off · caching

KEY MESSAGE Dependable federation needs maintainers, engine creators, and users working together.

[5] Existing starting points: VoID · SPARQL 1.1 Service Description

From silent failures to limit-aware federation.

Today: silent failures, errors, empty or unstable results → **Goal:** federation that knows and respects its limits

For more details see Discussion and Recommendations sections 6.3 & 8.

From silent failures to limit-aware federation.

Today: silent failures, errors, empty or unstable results → **Goal:** federation that knows and respects its limits

NEAR TERM



① Publish limits and status

Endpoints state their limits, e.g. **VoRD →**



② Make engines limit-aware

Engines plan requests within published limits

For more details see Discussion and Recommendations sections 6.3 & 8.

From silent failures to limit-aware federation.

Today: silent failures, errors, empty or unstable results → **Goal:** federation that knows and respects its limits

NEAR TERM



① Publish limits and status

Endpoints state their limits, e.g. **VoRD** →



② Make engines limit-aware

Engines plan requests within published limits

LONGER TERM



③ Evaluate in the wild, over time

Test on live endpoints, repeatedly over time



④ Sustainable infrastructure

Share the load: gateways, caches, **TPF**, etc.

KEY MESSAGE When endpoint limits are visible, engines can plan around them instead of failing.

For more details see Discussion and Recommendations sections 6.3 & 8.

Federation over public endpoints is not yet reliable.

Manual ≈ 21× more successful
96% of **algorithmic** executions failed

Both approaches declined
Endpoints moved, services retired, limits tightened

Visible limits are a way forward
Endpoints publish them (e.g. VoRD); engines adapt

RESOURCES

Explore the data
Results explorer ↗

Read the paper
Crum et al., ISWC 2026 ↗

Next: limits engines can read
VoRD ↗

SLIDES LINK

KEY MESSAGE Testing federation in the wild exposes what benchmarks miss: endpoint limits and change.

Remedies

Endpoint limits can be represented with VoRD, engines can start to consider them. Should frights linger, please reach out! We would be happy to try to help.

Supported by FWO · SolidLab Vlaanderen · Swiss National Science Foundation · EU Horizon Europe (MICROBES-4-CLIMATE) · Swiss Government SERI · NIH.

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Ruben Taelman • UGent

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fwo → SIB Swiss Institute of Bioinformatics

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GHENT UNIVERSITY

vito

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Bart Buelens • VITO

These slides were prepared with the help of Claude Opus 5.5 (Anthropic) in October 2026.

SUPPLEMENTARY SLIDES

Details and evidence

TERMS AND DEFINITIONS

S1 Terminology: talk and paper

S2 What counts as an answer

S3 SERVICE vs. no-service query

DETAILS BEHIND THE TALK

S4 Algorithmic approaches slide 03B

S5 How complex the queries are slide 04

S6 Error breakdown slide 07

S7 Request explosions slide 07

S8 Answers per run slide 08

S9 Endpoint changes slide 09, 10

S10 Queries affected by the EMI move slide 09

S11 Publishing endpoint limits (VoRD) slide 12

FURTHER MATERIAL

S12 Every query × every run

S13 The running example

S14 Limitations of the study

S15 References

Terms in this talk and in the paper.

IN THIS TALK	IN THE PAPER	WHAT IT MEANS
Manual federation	**SERVICE experiments** server-side manual federation	Query keeps its SERVICE clauses; one target endpoint federates and joins. 4 runs.
Algorithmic federation	**no-service experiments** client-side automated federation	SERVICE clauses removed; Comunica selects sources, plans and joins. 21 runs.
Federation engine	**Comunica v4.3.0** one client for both settings	Federates only in no-service runs; in SERVICE runs it just submits the query.
Algorithmic approaches	`NOMETA-ASK` · `NOMETA-COUNT` `VOID-TRIPLE` · `VOID-BLOCK` · `-NRL`	Comunica configurations for source selection and join planning (S4).
Run	**experiment** 25 experiments, Mar 2025 – Jun 2026	Every query of the workload executed once, with one approach and configuration.
Answered / success	**execution success rate** Sec. 4.4	Share of executions that return a non-empty result set.

Definitions from the paper, Sec. 4.3 (federation approaches) and 4.4 (experimental set-up).

What counts as an answer?

Non-empty result

Parseable SPARQL Results JSON with `result count > 0`.

Empty result

Parseable output, but zero bindings returned.

Explicit error

No usable output or an exception/error path.

Result-set size agreement is used as a proxy for consistency; the study does not claim semantic binding equality. Direct endpoint execution and the Comunica control were close (65.2% vs. 60.6% non-empty).

The running example, as SPARQL.

SERVICE form (manual)

`original/109_uniprot_transporter_in_liver.rq`

```
# Datasources: https://sparql.uniprot.org/sparql/
PREFIX genex: <http://purl.org/genex#>
PREFIX lscr: <http://purl.org/lscr#>
PREFIX obo: <http://purl.obolibrary.org/obo/>
PREFIX orth: <http://purl.org/net/orth#>
PREFIX rh: <http://rdf.rhea-db.org/>
PREFIX taxon: <http://purl.uniprot.org/taxonomy/>
PREFIX uberon: <http://purl.obolibrary.org/obo/uo#>
PREFIX up: <http://purl.uniprot.org/core/>
SELECT
  ?rhea
  ?protein
  ?anat
WHERE
{
  GRAPH <https://sparql.rhea-db.org/rhea> {
    ?rhea rh:isTransport true .
  }
  ?protein up:annotation ?ann .
  ?protein up:organism taxon:9606 .
  ?ann up:catalyticActivity ?ca .
  ?ca up:catalyzedReaction ?rhea .
  BIND(uberon:0002107 AS ?anat)
  SERVICE <https://www.bgee.org/sparql> {
    ?seq genex:isExpressedIn ?anat .
    ?seq lscr:xrefUniprot ?protein .
    ?seq orth:organism ?organism .
    ?organism obo:RO_0002162 taxon:9606 .
  }
}
```

no-service form (algorithmic)

`no-service/109_uniprot_transporter_in_liver_ns.rq`

```
# Datasources: https://sparql.uniprot.org/sparql/ https://www.bgee.org/sparql
PREFIX genex: <http://purl.org/genex#>
PREFIX lscr: <http://purl.org/lscr#>
PREFIX obo: <http://purl.obolibrary.org/obo/>
PREFIX orth: <http://purl.org/net/orth#>
PREFIX rh: <http://rdf.rhea-db.org/>
PREFIX taxon: <http://purl.uniprot.org/taxonomy/>
PREFIX uberon: <http://purl.obolibrary.org/obo/uo#>
PREFIX up: <http://purl.uniprot.org/core/>
SELECT
  ?rhea
  ?protein
  ?anat
WHERE
{
  GRAPH <https://sparql.rhea-db.org/rhea> {
    ?rhea rh:isTransport true .
  }
  ?protein up:annotation ?ann .
  ?protein up:organism taxon:9606 .
  ?ann up:catalyticActivity ?ca .
  ?ca up:catalyzedReaction ?rhea .
  BIND(uberon:0002107 AS ?anat)
  {
    ?seq genex:isExpressedIn ?anat .
    ?seq lscr:xrefUniprot ?protein .
    ?seq orth:organism ?organism .
    ?organism obo:RO_0002162 taxon:9606 .
  }
}
```

One change: the `SERVICE <https://www.bgee.org/sparql>` wrapper becomes a plain group and Bgee joins the data sources, so the engine must place every pattern itself. Files verbatim from fed-survey-results/queries ↗.

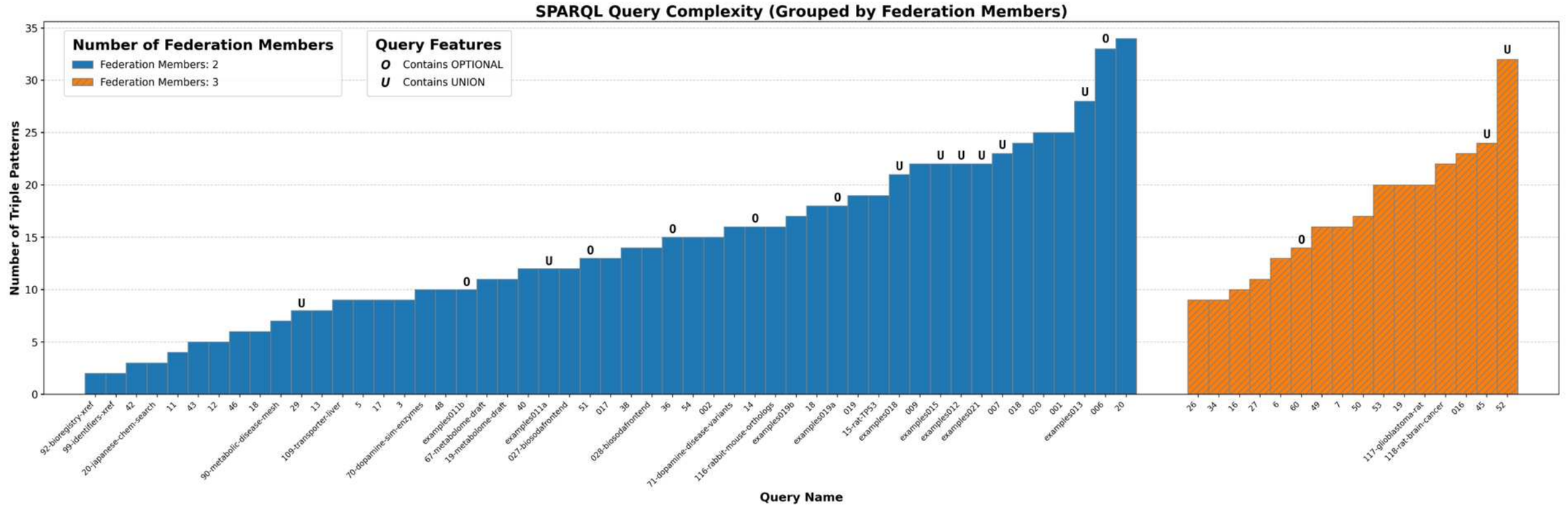
All algorithmic approaches showed low success rates.

CONFIGURATION	WHAT IT USES TO SELECT SOURCES AND PLAN	RUNS	NON-EMPTY RESULTS
NOMETA-ASK live ASK probes	✓ ASK · COUNT · VoID · Large est. · ✓ Rate limit	8 runs incl. 3 -NRL	12 / 480 · 2.50%
NOMETA-COUNT live COUNT probes	ASK · ✓ COUNT · VoID · Large est. · ✓ Rate limit	5 runs incl. 1 -NRL	5 / 300 · 1.67%
VOID-TRIPLE VoID, per triple pattern	✓ ASK · COUNT · ✓ VoID · Large est. · ✓ Rate limit	4 runs	6 / 240 · 2.50%
VOID-BLOCK VoID, larger query blocks	✓ ASK · COUNT · ✓ VoID · ✓ Large est. · ✓ Rate limit	4 runs	6 / 240 · 2.50%
Manual (SERVICE) for comparison	the target endpoint follows the SERVICE clauses	4 runs	123 / 260 · 47.3%

1.67–2.50% for every configuration. VoID metadata helped only slightly; turning off client rate limiting (-NRL) made no clear difference.

Configurations and results from the paper, Sec. 4.3 (Table 1) and 5.4; all ran in Comunica 4.3.0. Manual figure from the 25-run results explorer data.

Moderate syntax, non-trivial federation.



LIVE ↗ 2 federation members 3 federation members **O** OPTIONAL · **U** UNION Hover a bar for details · **Open full screen ↗**

Average: 14.9 triple patterns (2–34) · 2.24 federation members · few `OPTIONAL` / `UNION` .

The errors had an operational signature.

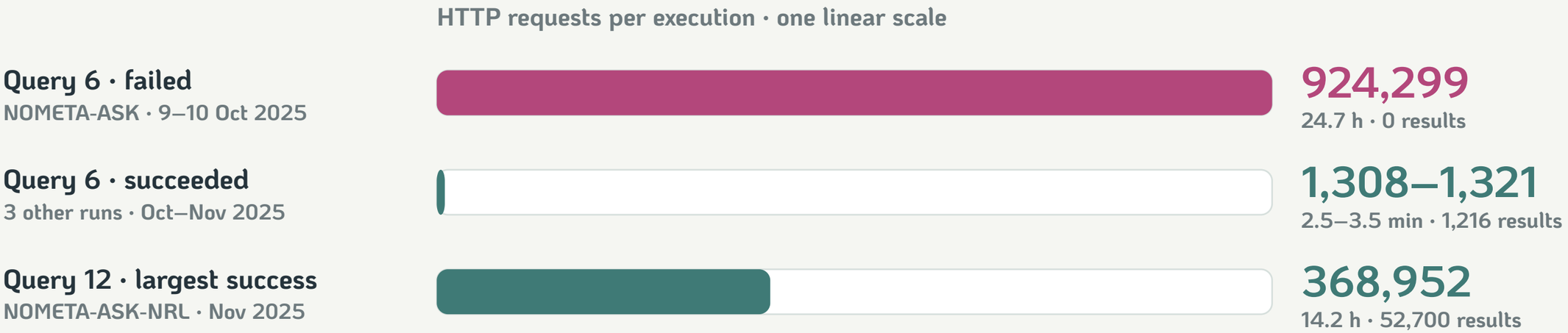
ERROR GROUP	EXECUTIONS ▪ algorithmic ▪ manual	WHAT IT LOOKS LIKE	MEDIAN REQ.
Client-side HTTP 4xx	850 814 / 36	Endpoint refused the request: mostly 403 Forbidden, some 400/404	5 requests
Server-side HTTP 5xx	245 216 / 29	Endpoint failed while answering (HTTP 500 and other 5xx)	4 requests
Transport-level network	91 91 / 0	Connection dropped mid-transfer: “fetch failed”, request failed	2,004 requests
Other malformed	90 53 / 37	HTML error page sent as HTTP 200, parse errors, client out of memory	78 requests
Timeout-related long runs	63 39 / 24	Long runs aborted or terminated before finishing	699 requests

82% were 4xx/5xx from endpoints, usually within ~5 requests: the queries were not faulty.

Error heatmap ↗

1,339 errors in 1,520 executions (results explorer); the paper (Sec. 5.2) lists 242 / 82 / 102 for 5xx / transport / other, before 12 records were reclassified.

Query 6: 1,308 requests to succeed, 924,299 to fail.



What the 924,299 requests of the failed run were

One lookup per binding

303,503 bind-join steps, each a small (~170-byte) request carrying one bound value.

Sent to all three endpoints

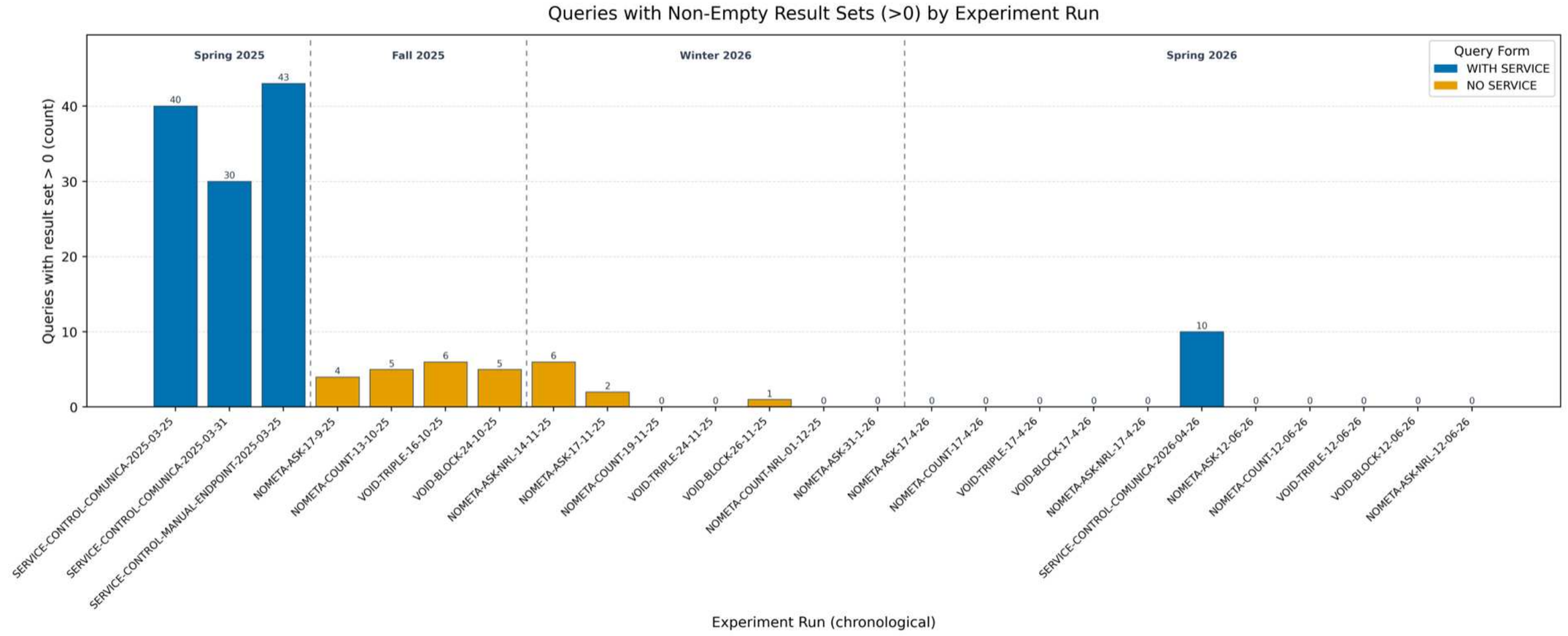
303,503 each to SwissLipids and Rhea; 317,293 to UniProt, incl. 13,790 retries after “429 Too Many Requests”.

Steady for a full day

About 37,000 requests per hour for 24.7 h, until the connection failed (“fetch failed”).

Successful query 6 runs sent 13 ASK probes per endpoint, then queried only Rhea and UniProt; the failed run did not log request bodies (results explorer ↗).
Query 6 was not alone: in the same NOMETA-ASK run (Sep–Oct 2025), queries 13 and 16 also failed, after 742,417 and 553,202 requests.

Non-empty result sets for each of the 25 runs.



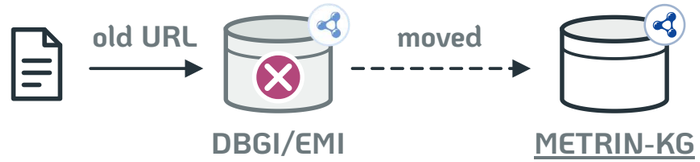
LIVE ↗ SERVICE (manual) no-service (algorithmic)

Hover a bar for details · Open in explorer ↗

Queries with result set > 0, per experiment run (dashboard order); chronological version in the paper (Fig. 2).

A live endpoint is part of the experimental treatment.

WHAT CHANGED

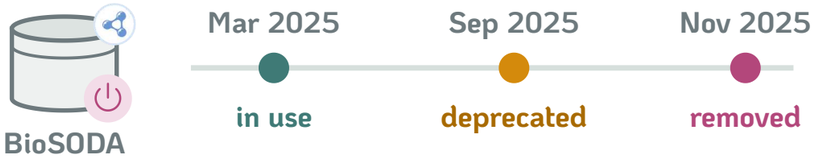


WHY QUERIES BROKE

Queries still named the old URL; clients got no machine-readable notice, so the requests failed.

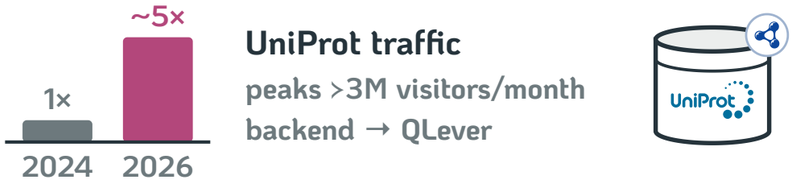
WHAT WOULD HELP

Announce moves
redirects & migration signals



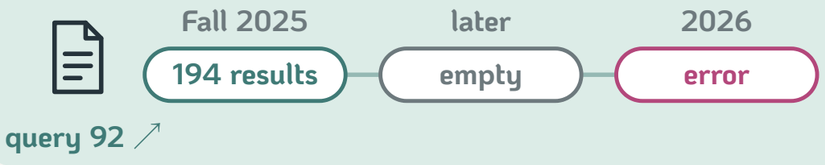
14 of 67 queries federated through this temporary server; once it was removed, they could only fail.

Publish service status
status & deprecation notices



Stricter limits on response time, concurrency, request rate and volume: request-heavy plans got **HTTP 403**.

Publish and respect limits
VoRD (S11); budgets & back-off



2 in 3 queries varied in result count.
Data releases (UniProt: 3, +28B triples), endpoint changes and failures.

Keep query records
query, sources, results (**QVMC**)
e.g. **Solid Cockpit** with Solid Pods

From the paper: changes Sec. 5.1, 5.5 and 6.2; recommendations Sec. 6.3 and 8. Query records: QVMC specification, implemented in Solid Cockpit.

14 queries depended on one endpoint that moved.

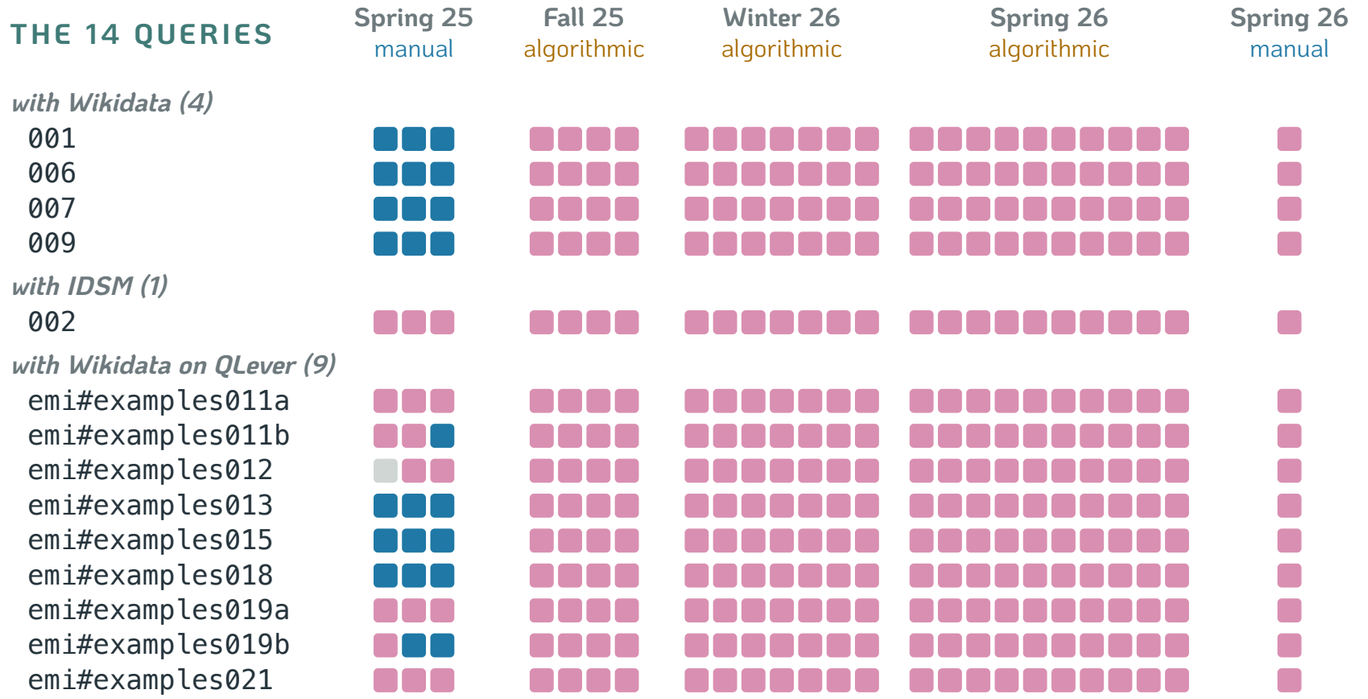
EMI knowledge graph (DBGI) on the BioSODA server
`biosoda.unil.ch/emi/sparql`

moved

METRIN-KG
`kg.earthmetabolome.org/metrin`

Mar 2025 — in use
Sep 2025 — deprecated
Nov 2025 — removed

THE 14 QUERIES



9 of 14 answered in Spring 2025 · **0 of 14** in any later run

NEXT: RE-ASSESS THESE QUERIES

1. **Retarget the queries**
   swap biosoda.unil.ch/emi/sparql for **METRIN-KG** in both query forms
2. **Re-check positive controls**
   run each query in the endpoint's web interface, as in March 2025
3. **Re-run both approaches**
   SERVICE and the algorithmic configurations, as in the study
4. **Compare with the baseline**
   9 of 14 answered in Spring 2025; are the answers the same?

Queries and targets: SIB query corpus; outcomes: 25-run results explorer; changes: paper Sec. 5.5. The re-assessment is planned future work, not part of the paper.
METRIN-KG endpoint as documented in Tandon et al., GigaScience 2026.

VoRD: machine-readable descriptions of endpoint limits.



Endpoint maintainer

describes its limits once, next to its SPARQL endpoint

query.wikidata.org/sparql — VoRD description ↗

Anonymous clients
≤ 10 requests / min → HTTP 429 + Retry-After

With a User-Agent
≤ 200 requests / min

Established users
≤ 2,000 requests / min

Parallel requests
≤ 3 at once per IP (guidance) → HTTP 503

```
ex:anonymous a vord:RateRestriction ;
vord:maxValue 10 ; vord:windowDuration "PT1M" .
```

Federation engines
plan requests within the limits and adapt before reaching them

Users
know the limits before querying, and why a request was rejected

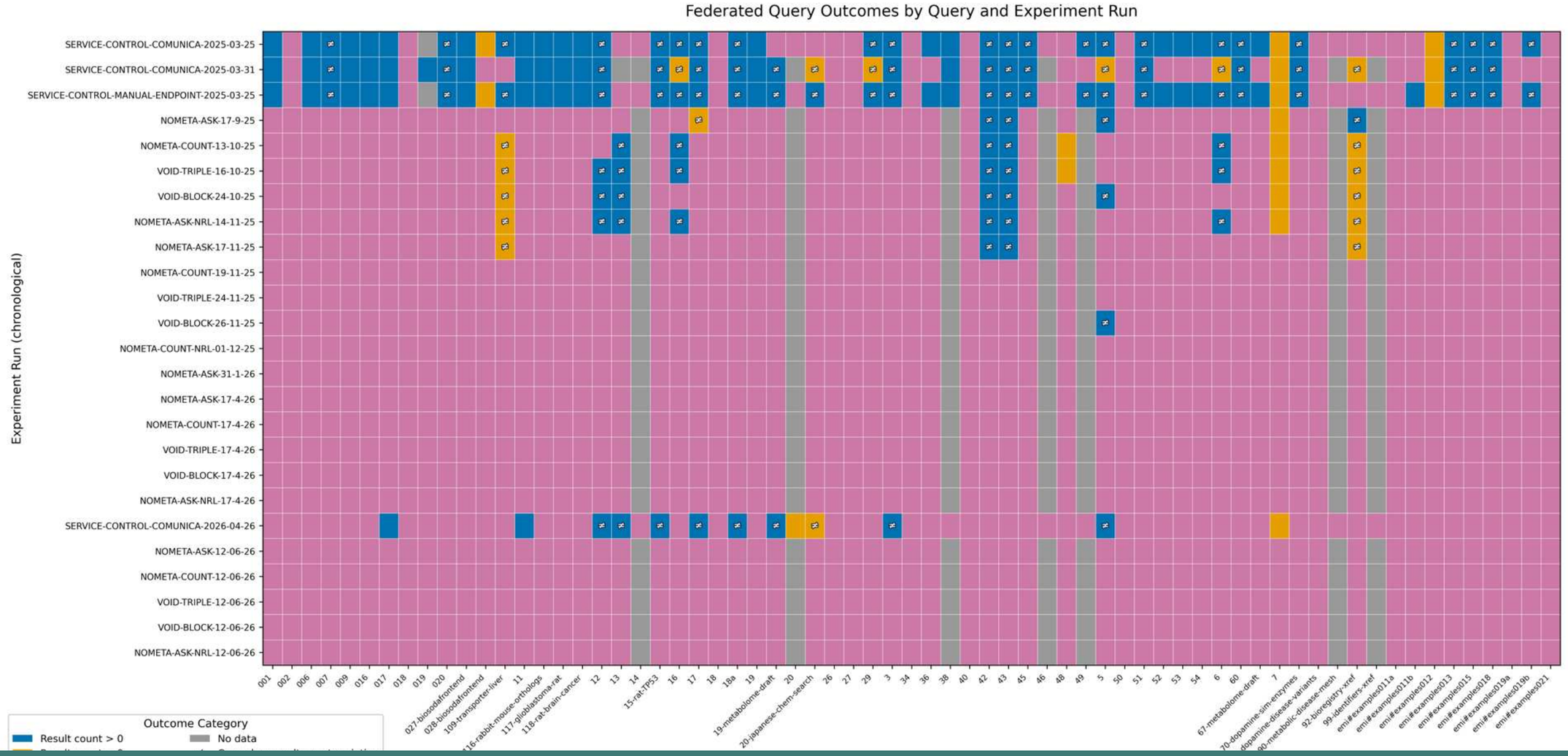
Without published limits: the cycle from slide 10
traffic → tighter limits → failed queries → retries → more traffic

With published limits, the cycle breaks
plan within limits → fewer rejections, fewer retries

KEY MESSAGE If endpoints publish their limits, engines and users can stay within them.

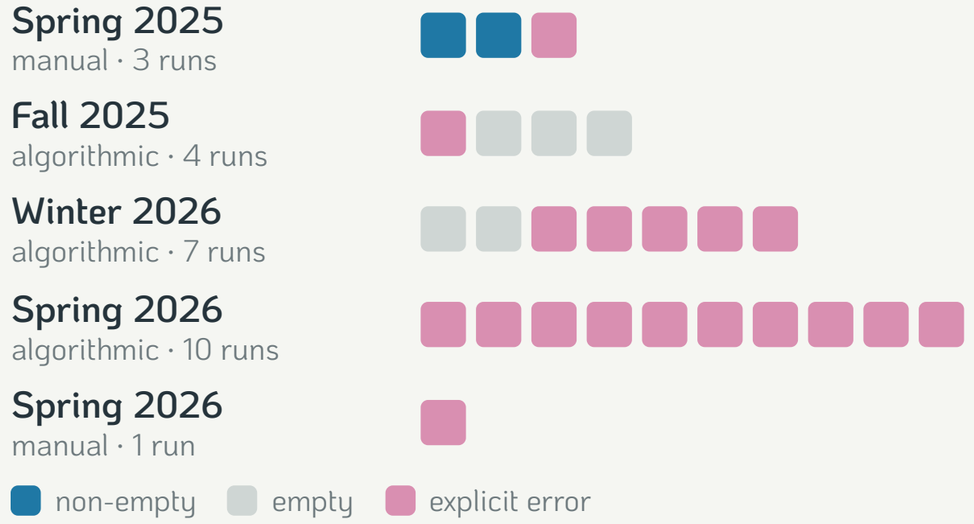
VoRD (v0.2) extends SPARQL 1.1 Service Description and is validated with SHACL and ShEx.
Card values: VoRD's Wikidata example, derived from Wikimedia's public rate-limit documentation (May 2026).

Failures were widespread across the query set.



Query 109: transporter proteins in the liver.

```
# sent to sparql.uniprot.org (PREFIXes omitted)
SELECT ?rhea ?protein ?anat
WHERE {
  GRAPH <https://sparql.rhea-db.org/rhea> {
    ?rhea rh:isTransport true .
  }
  ?protein up:annotation ?ann .
  ?protein up:organism taxon:9606 .
  ?ann up:catalyticActivity ?ca .
  ?ca up:catalyzedReaction ?rhea .
  BIND(uberon:0002107 AS ?anat)
  SERVICE <https://www.bgee.org/sparql> {
    ?seq genex:isExpressedIn ?anat .
    ?seq lscr:xrefUniprot ?protein .
    ?seq orth:organism ?organism .
    ?organism obo:RO_0002162 taxon:9606 .
  }
}
```



Spring 2025: one manual run hit a malformed response (parse error).

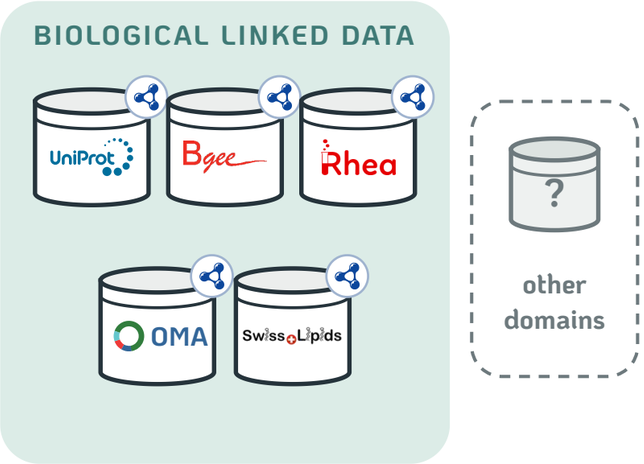
Fall 2025: empty results (~21 requests); one run aborted after 609 requests.

From Nov 2025: HTTP 403 Forbidden within 2–12 requests; UniProt (and Bgee, Jun 2026) rejected the engine's ASK probes.

Query: UniProt SPARQL example `109_uniprot_transporter_in_liver`, federating UniProt and Bgee. Per-run logs: fed-survey-results ↗.
The paper's Figure 1 shows a different query, *query 7* (SwissLipids), which never returned a non-empty result.

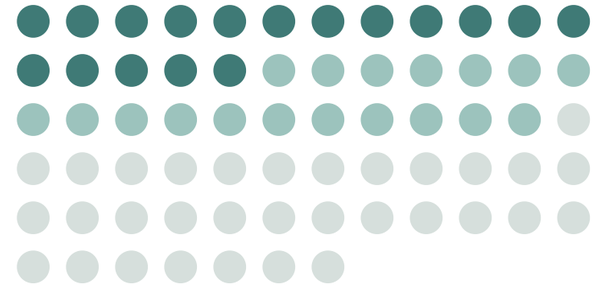
What this study does, and does not, claim.

01 Domain-specific



Workload from users of SIB-hosted biological endpoints (7 target, 14 federated); other domains may differ.

02 Meaningful, not exhaustive



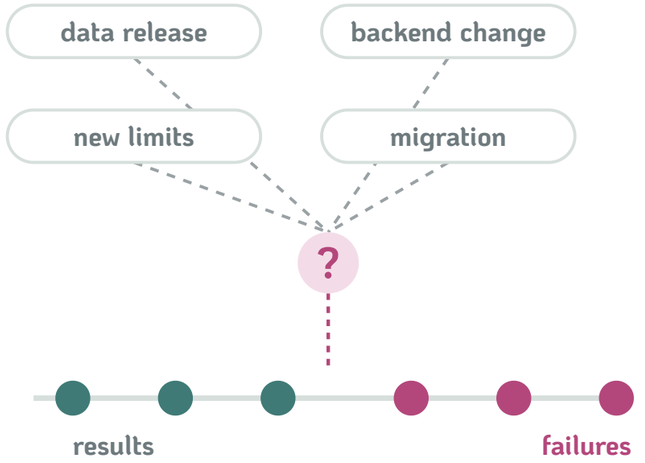
~¼ led to published results

together >½ with a real-world use

avg. 14.9 triple patterns · 2.24 sources

67 expert-written queries: realistic evidence of practice, not a universal benchmark.

03 Causality is incomplete



Logs show when an outcome changed, not always which change caused it.

What we do show: on live public biological endpoints, federation was unreliable and degraded over time, for both approaches.

From the paper: workload Sec. 4.2, limitations Sec. 7.

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16. **Crum, E.** VoRD: Vocabulary of Restrictive Datasets, v0.2.